A FIELD GUIDE FOR ML IN FRAUD & CREDIT

A fraud model can look perfect and still be broken.

10 silent failure modes — and the checks that catch them.

I validated my own fraud model the way a bank's Model Risk team would. Six of these ten failures were live in it — the red-tagged slides are the ones I caught in my own build.

Save this for your next model review →

FAILURE MODE 01

Leakage by resampling

SMOTE or scaling applied before the train/test split; random splits leak future fraud-ring members into training.

THE CHECK

Audit pipeline order. Time-ordered splits. Point-in-time features only.

FAILURE MODE 02

The delayed-label blind period

Fraud labels arrive 1–6 months late. The model decays silently while dashboards show stale wins.

THE CHECK

Annotate label lag on every metric. Use early proxy signals (disputes, soft labels).

FAILURE MODE 03

Selective-labels feedback loop

Only investigated cases get labels — the model retrains on its own past decisions and calcifies its blind spots.

THE CHECK

Sample and label below the threshold. Validator-side label audits.

FAILURE MODE 04 · CAUGHT IN MY OWN MODEL

Adversarial adaptation

Fraud is a cat-and-mouse game. Fraudsters probe thresholds and route around static models.

THE CHECK

Perturbation-test manipulable inputs. (Mine: –20% stated income flipped 92% of the queue.)

FAILURE MODE 05 · CAUGHT IN MY OWN MODEL

Hidden single point of failure

One feature family quietly carries the model. An outage or evasion of it collapses detection.

THE CHECK

Family ablation. (Mine: PR-AUC 0.87 → 0.47 without affordability ratios.)

FAILURE MODE 06 · CAUGHT IN MY OWN MODEL

Unearned complexity

The tuned champion ships because it's the champion. Nobody asks if simpler matches it.

THE CHECK

Challenger benchmark. (Mine: the challenger won, 0.887 vs 0.873.)

FAILURE MODE 07 · CAUGHT IN MY OWN MODEL

Segment blind spots

Aggregate metrics look great while a channel or fraud type is under-detected.

THE CHECK

Slice everything: fraud type, channel, amount band. (Mine: 90% ring vs 57% solo capture.)

FAILURE MODE 08 · CAUGHT IN MY OWN MODEL

Drift-alarm fatigue

Uniform PSI thresholds on bursty features alarm daily — until the team ignores all alerts.

THE CHECK

Calibrate thresholds per feature family; monitor score PSI first. (Mine: 9 features 'drifting', score stable.)

FAILURE MODE 09 · CAUGHT IN MY OWN MODEL

Proxy discrimination

Age or postcode enter as innocent features — and become indirect discrimination the day scores gate decisions.

THE CHECK

Attribution review on protected attributes + a fairness gate before scope expands. (Mine: age ranked top-6 by SHAP.)

FAILURE MODE 10

Scores that aren't probabilities

Rank-good, probability-bad scores silently corrupt expected-loss maths and risk thresholds.

THE CHECK

Reliability curves and Brier on the out-of-time fold — not the training set.

THE META-LESSON

7 of 10 are invisible to the standard workflow: train → test → report AUC → ship.

They surface only under independent validation — which is why regulators have required it since SR 11-7 (2011), and why AI validation is one of the most underrated career paths in data science right now.

Full validation report, findings log and reproducible code: github.com/Kriti-Data-Business

Part 1 of this series: the Two Hats story. Part 3: validating an LLM assistant. Follow for the rest.